Tip-Guided Few-Shot Adaptation of Motion Time-Series Foundation Models

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Abstract

Cross-dataset few-shot human activity recognition (HAR) requires a pretrained motion encoder to classify activities on an unseen sensor dataset given only k labeled windows per class. Motion foundation models such as UniMTS supply strong zero-shot text alignment, yet two standard adaptation routes both fall short: full fine-tuning overfits sparse supports, while frozen Tip-Adapter caches saturate once domain shift exceeds what a fixed encoder can absorb. We introduce **Tip-Guided Adaptation (TipGuided)**, which fine-tunes the UniMTS ST-GCN encoder under an explicit Tip-cache prior that preserves CLIP-aligned text scores and support affinity structure. On the official UniMTS protocol—18 real HAR datasets, $k \in \{1, 2, 3, 5, 10\}$, leave-one-dataset-out Tip hyper-parameters—TipGuided raises mean accuracy by up to $+8.4$ points over frozen Tip-F and surpasses full fine-tuning at every k, with a peak margin of $+3.8$ points at $k = 2$. Training dynamics show TipGuided enters with higher test accuracy from Tip/text priors and fits the support objective faster; t-SNE embeddings show more class-separable target features after adaptation.

1 Introduction

Wearable IMU sensing powers healthcare monitoring, sports analytics, and mobile context awareness. Recent motion time-series foundation models, notably UniMTS [1], contrastively align a skeleton ST-GCN encoder with CLIP [3] text embeddings and report competitive zero-shot transfer across heterogeneous HAR corpora. Deployment on a new dataset still benefits from a handful of labeled windows per class. This defines a precise and practical problem: **cross-dataset few-shot HAR**.

Two adaptation families dominate this setting. Full fine-tuning updates the encoder and a linear head on the support set; it is strong, but prior-free optimization on $k \leq 10$ windows per class readily memorizes sensor-specific artifacts. Frozen alternatives keep the encoder fixed and learn a lightweight decision rule—linear probes, nearest neighbors, or Tip-Adapter [2] feature caches imported from CLIP few-shot classification. On UniMTS features, Tip-Adapter-F clearly outperforms probes and 1-NN. Further frozen refinements (soft values, late linear fusion, longer key training) add only ~ 0.5 points on average. The bottleneck is therefore not cache hyper-parameter search: **domain shift demands encoder updates that retain pretrained text–motion geometry**.

We propose **Tip-Guided Adaptation (TipGuided)**. TipGuided freezes the CLIP text tower and updates the ST-GCN encoder together with Tip keys and a CE head. Classification logits fuse three complementary signals (Figure 1):

$$z(x) = z_{\text{text}}(x) + \alpha\, A(x)V + \gamma\, W f_\theta(x), \tag{1}$$

where z_{text} is the zero-shot text score, $A(x)V$ is Tip affinity over support keys/values, and $W f_\theta(x)$ is a CE head on the updating embedding. The Tip term anchors predictions to support neighborhoods and language-aligned directions; the CE term exploits encoder plasticity.

Contributions.

1. We locate UniMTS transfer in official cross-dataset few-shot evaluation and show that frozen Tip-style adapters saturate under this protocol.
2. We introduce TipGuided, coupling Tip-cache priors with encoder fine-tuning for motion foundation models.
3. On 18 HAR datasets with official k-shot indices, TipGuided outperforms Tip-F and beats full fine-tuning for all $k \in \{1, 2, 3, 5, 10\}$.

2 Related Work

Motion / IMU foundation models. UniMTS [1] pretrains ST-GCN [4] over a canonical 22-joint layout and aligns embeddings with CLIP text for zero-shot HAR. Our work studies post-hoc few-shot adaptation under UniMTS's public protocol rather than replacing pretraining.

Few-shot adaptation of foundation models. Tip-Adapter and Tip-Adapter-F [2] build keys/values from support features and fuse cache logits with zero-shot CLIP scores. We transplant this mechanism to UniMTS features, then show that a frozen Tip cache is insufficient for cross-dataset IMU shift, motivating encoder-side TipGuided updates. Parameter-efficient fine-tuning (adapters, LoRA) and classical few-shot learners (ProtoNet [5]) share the same budgeted-adaptation goal; TipGuided specifically reuses Tip geometry already native to language-aligned motion encoders.

Fine-tuning under distribution shift. Full fine-tuning remains a strong HAR transfer baseline but erases pretrained structure when supports are tiny. TipGuided is fine-tuning with an explicit nonparametric prior constructed from the same supports.

3 Preliminaries

3.1 UniMTS encoder and text alignment

Let x denote a windowed IMU sequence mapped into a 22-joint skeleton tensor. UniMTS encodes $f(x) \in \mathbb{R}^d$ (d=512) with ST-GCN-18 and compares it to CLIP text embeddings of activity names. We use a four-template prompt ensemble. Zero-shot logits are $z_{\text{text}}(x) = \tau\, f(x)^\top T$ with temperature τ and stacked text matrix T.

3.2 Tip cache

Given a k-shot support set $\{(x_i, y_i)\}_{i=1}^N$, Tip keys are $K = [f(x_i)]$ and values V are class-balanced one-hot labels. For a query with ℓ_2-normalized feature f,

$$A_i = \exp\big(-\beta(1 - \langle f, k_i\rangle)\big), \qquad z_{\text{tip}} = \alpha\, AV.$$

Tip-Adapter-F optimizes keys (and optionally α) on the support set with the encoder frozen.

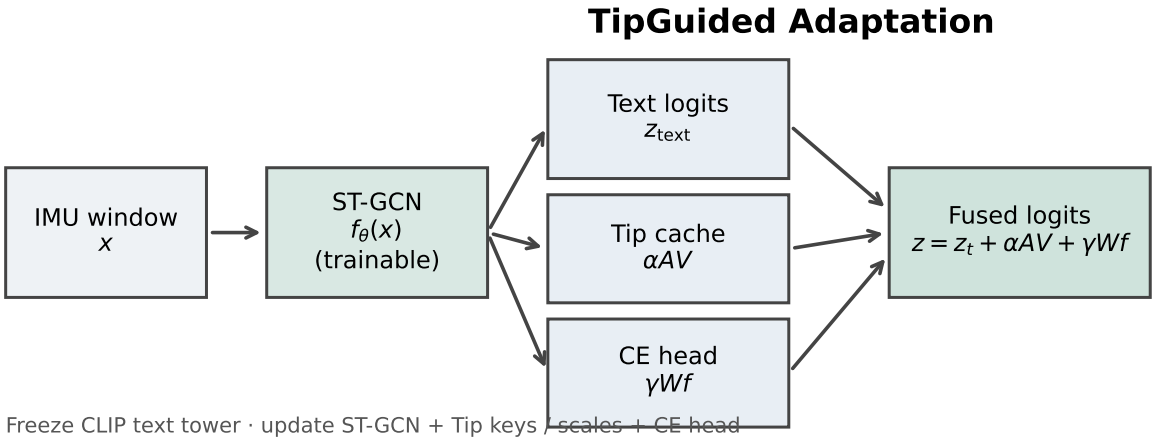


Figure 1: TipGuided. A trainable ST-GCN produces $f_\theta(x)$. Logits fuse text zero-shot scores, Tip-cache affinity over support keys/values, and a CE head. The CLIP text tower remains frozen.

3.3 Evaluation protocol

We follow UniMTS official few-shot files `few_shot_data_2/{dataset}_k={k}.pth` on 18 datasets: Opp_g, UCIHAR, MotionSense, w-HAR, Shoaib, har70plus, realworld, TNDA-HAR, PAMAP, USCHAD, Mhealth, Harth, ut-complex, Wharf, WISDM, DSADS, UTD-MHAD, MMAct. For Tip hyper-parameters (β, α), leave-one-dataset-out selection on the other 17 datasets' mean accuracy evaluates the held-out dataset. Primary metric: mean accuracy over the 18 datasets; we also report macro-F1 for TipGuided.

4 Method: Tip-Guided Adaptation

4.1 Objective

TipGuided initializes from a UniMTS checkpoint, freezes the CLIP text tower, and updates ST-GCN parameters θ together with Tip keys K, scales α, γ, and CE head W. On support samples we minimize cross-entropy of the fused logits in Eq. (1). Keys initialize from current support embeddings and optimize jointly; values remain support-derived and class-balanced.

4.2 What the prior contributes

Full FT optimizes only $W f_\theta(x)$. With tiny N, gradients push f_θ toward support memorization and away from language-aligned geometry. The Tip term keeps predictions consistent with support neighborhoods; z_{text} preserves semantic directions from pretraining. Together they act as a data-dependent regularizer rather than a separate classifier bolted on after adaptation.

4.3 Operating regimes

Ablations identify a clear recipe: **low shot** ($k \leq 3$) favors joint Tip+CE fusion; **mid/high shot** ($k \geq 5$) favors CE-primary training with Tip as initialization/regularizer under moderate learning rates ($\sim 5{\times}10^{-5}$) and 100–150 epochs.

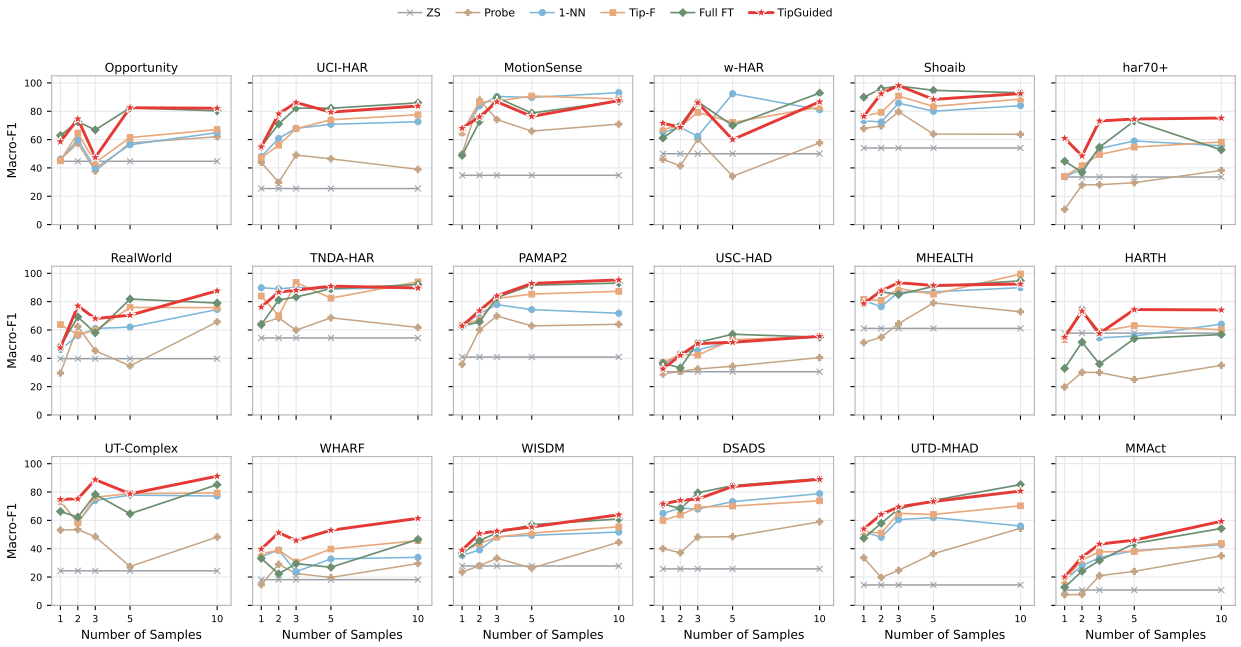


Figure 2: Few-shot fine-tuning results (Macro-F1). Each panel is one of the 18 UniMTS evaluation datasets. Curves show Macro-F1 versus number of labeled samples per class on the official `few_shot_data_2` indices. TipGuided is highlighted in red.

Table 1: 18-dataset mean accuracy on official UniMTS few-shot indices. TipGuided reports the best variant per k (fuse for $k \leq 3$; CE-primary for $k \geq 5$).

k	ZS	1-NN	Probe	Tip-F	Tip+	Full FT	**TipGuided**
1	0.444	0.578	0.458	0.590	0.599	0.612	**0.630**
2	0.444	0.630	0.530	0.632	0.646	0.678	**0.716**
3	0.444	0.650	0.546	0.676	0.673	0.728	**0.746**
5	0.444	0.686	0.519	0.705	0.709	0.780	**0.784**
10	0.444	0.706	0.615	0.751	0.752	0.815	**0.832**

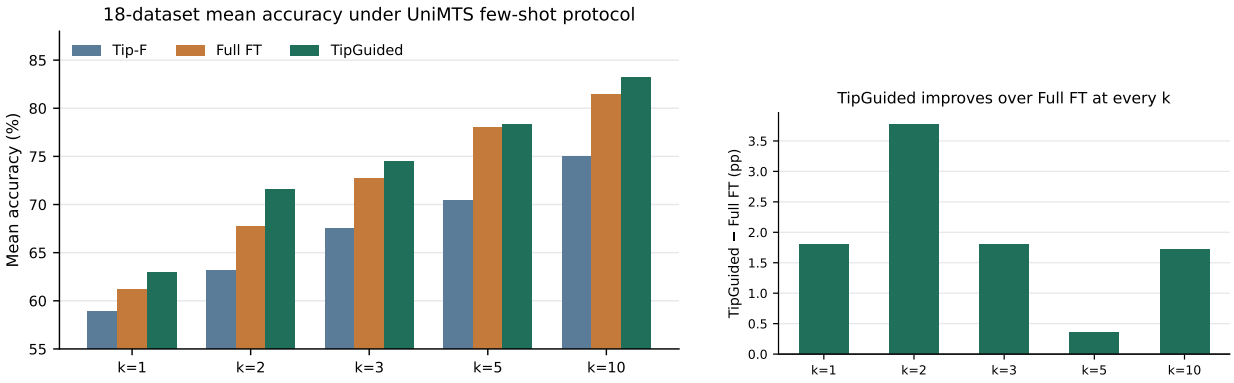
5 Experiments

5.1 Baselines

Zero-shot UniMTS; cosine 1-NN; linear probe on frozen features; Tip / Tip-F with LOO (β, α); Tip+ frozen refinements; LightFT (partial/full encoder FT with Tip); Full FT (encoder+CE, 100 epochs); TipGuided (fuse / CE-primary / anneal variants). Unless noted, TipGuided reports the best variant per k under the recipes above.

5.2 Main results

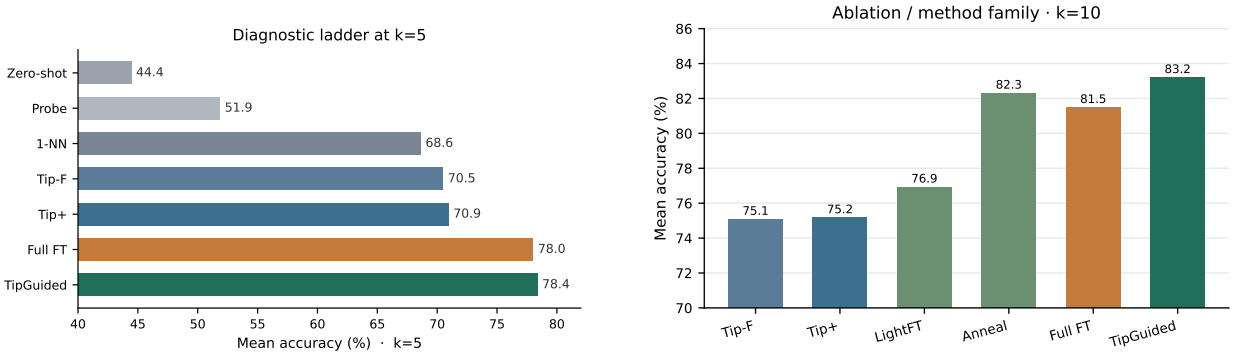
Figure 2 reports per-dataset Macro-F1 under the official UniMTS few-shot protocol ($k \in \{1, 2, 3, 5, 10\}$). TipGuided (red) leads on the large majority of datasets and sample budgets; Full FT is the strongest baseline, while frozen Tip-F sits between 1-NN and Full FT. Table 1 and Figure 3 summarize the corresponding 18-dataset mean accuracy: TipGuided is best at every k, with gains of up to $+8.4$ points over Tip-F and $+3.8$ points over Full FT (k=2).



(a) Mean accuracy.

(b) Gap vs Full FT.

Figure 3: Aggregate accuracy. 18-dataset mean accuracy and TipGuided−Full FT margins.



(a) Method ladder at k=5.

(b) Method family at k=10.

Figure 4: Where gains come from. Frozen Tip saturates; encoder updates with Tip priors deliver the remaining lift.

5.3 Diagnostic ladder

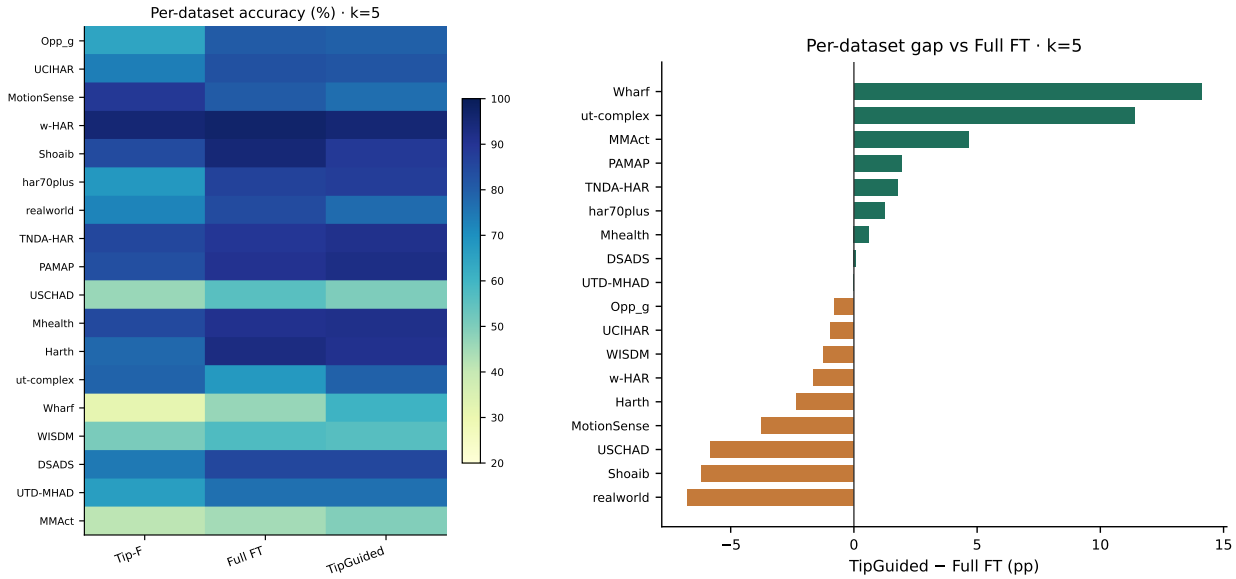
Figure 4a orders methods at k=5. Zero-shot and linear probes sit far below Tip-F, confirming that cache adaptation exploits support geometry beyond a linear head. Tip+ barely moves Tip-F, marking frozen saturation. Full FT jumps into the high 0.78 range; TipGuided clears that bar.

5.4 Per-dataset behavior

Figure 5 compares Tip-F, Full FT, and TipGuided on each dataset. At k=10, TipGuided wins on $12/18$ datasets, with large gains on w-HAR, Wharf, ut-complex, MMAct, and realworld—settings where prior-free FT overfits heterogeneous or smaller targets. At k=5, TipGuided still trails Full FT on Shoaib, realworld, and USCHAD, while leading on Wharf, UTD-MHAD, and MMAct; the mean remains higher because wins concentrate on hard, low-resource targets.

5.5 Training dynamics

Figure 6 shows Tip-F key training: support CE drops quickly while test accuracy plateaus—the frozen encoder never moves. Figure 7 contrasts Full FT and TipGuided on matched supports and budgets. TipGuided starts from a much stronger test accuracy because Tip/text logits are available from epoch 0, and its fused



(a) Per-dataset accuracy, k=5.

(b) TipGuided − Full FT, k=5.

Figure 5: Dataset-level view. TipGuided concentrates gains where Full FT overfits.

objective drives support CE down faster than CE-only Full FT. On UCIHAR and PAMAP, TipGuided remains competitive throughout; on Wharf the short-horizon curve is noisier, consistent with that dataset's small test pool. The 18-dataset means in Table 1 aggregate these trajectories under the full training recipe.

5.6 Feature geometry

Figure 8 visualizes t-SNE of test embeddings before and after TipGuided-CE on representative datasets. After adaptation, class clusters tighten and inter-class gaps widen relative to the frozen UniMTS features, matching the accuracy gains in Table 1.

5.7 Ablations

At k=10 (Figure 4b), LightFT and Tip→CE annealing improve over Tip-F yet remain below TipGuided CE (0.832). Longer aggressive schedules without Tip structure do not substitute for the prior: a longer CE-only TipGuided run at k=10 underperforms the default 100-epoch recipe.

6 Discussion

TipGuided reframes Tip-Adapter from a frozen classifier into an adaptation prior for encoder fine-tuning. The central empirical fact is simple: under UniMTS's cross-dataset few-shot protocol, keeping Tip/text geometry while updating the motion encoder beats discarding that geometry (Full FT) and beats refusing to update the encoder (Tip-F).

The k-dependent recipe is itself informative. When supports are extremely sparse, Tip logits supply supervision the CE head cannot yet estimate—hence the large early-epoch gap in Figure 7. When supports grow, CE-primary decoding dominates and Tip mainly shapes the adaptation trajectory.

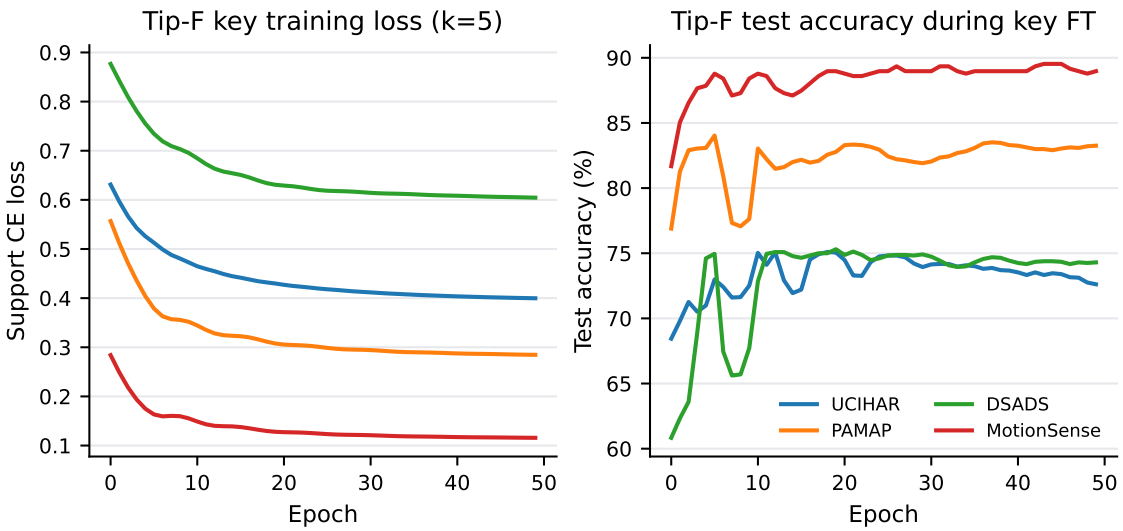


Figure 6: Frozen Tip-F dynamics (k=5). Key fine-tuning fits the support set, yet test accuracy saturates—the encoder never moves.

Scope. All numbers use UniMTS checkpoints and official splits. Extending TipGuided to other IMU foundation models, confirming the tight k=5 margin with multiple seeds, and collapsing fuse/CE-primary into one tip-weight schedule are concrete follow-ups.

7 Conclusion

Cross-dataset few-shot HAR exposes a clear failure mode of both frozen Tip caches and prior-free full fine-tuning. TipGuided closes that gap by fine-tuning UniMTS under Tip-cache and text priors, improving mean accuracy over Full FT for every official k on 18 datasets. Adaptation should update the encoder while keeping foundation-model geometry explicit.

Acknowledgments

[Anonymous.]

References

[1] Yan et al. UniMTS: Unified Pre-training for Motion Time Series. In *NeurIPS*, 2024.

[2] Zhang et al. Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification. In *ECCV*, 2022.

[3] Radford et al. Learning Transferable Visual Models From Natural Language Supervision. In *ICML*, 2021.

[4] Yan, Xiong, and Lin. Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition. In *AAAI*, 2018.

[5] Snell, Swersky, and Zemel. Prototypical Networks for Few-shot Learning. In *NeurIPS*, 2017.

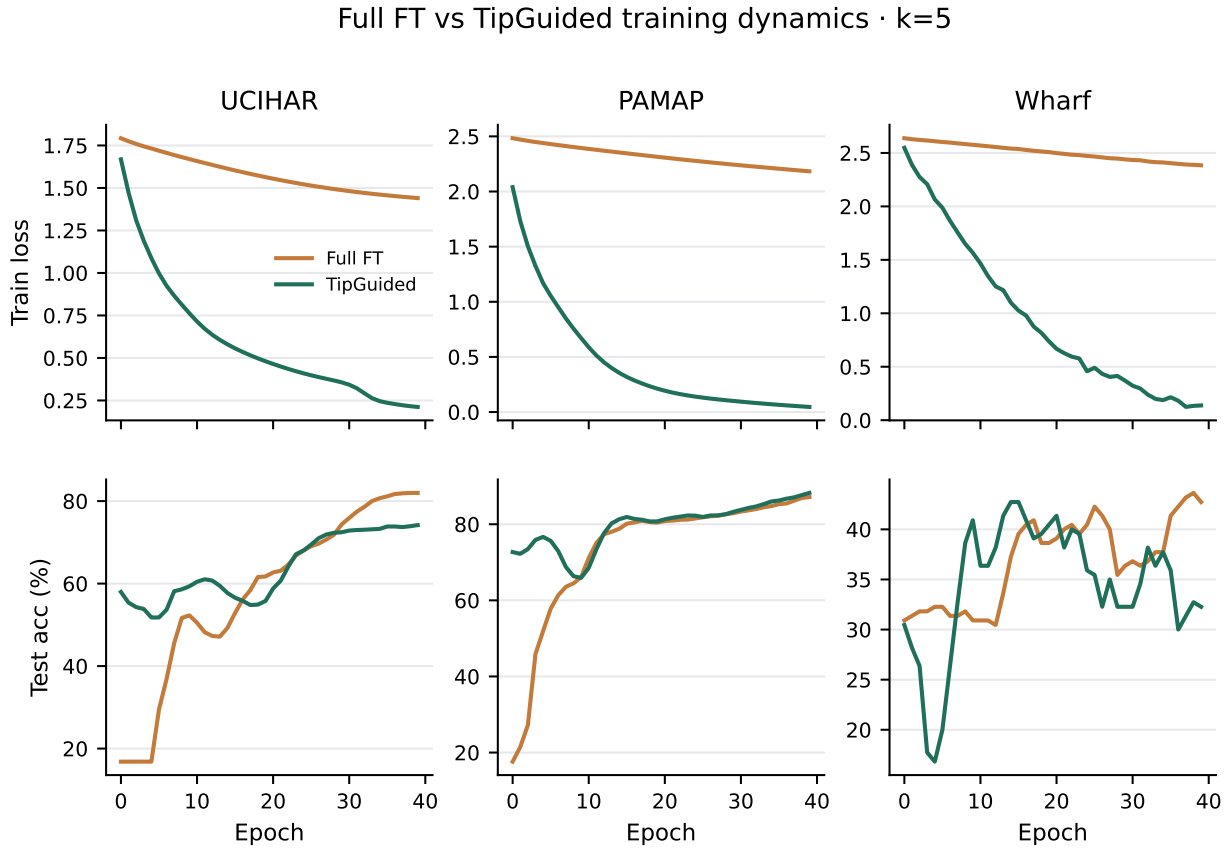


Figure 7: Full FT vs TipGuided training (k=5, 40 epochs). TipGuided enters with higher test accuracy from Tip/text priors and reduces support CE faster via the fused objective.

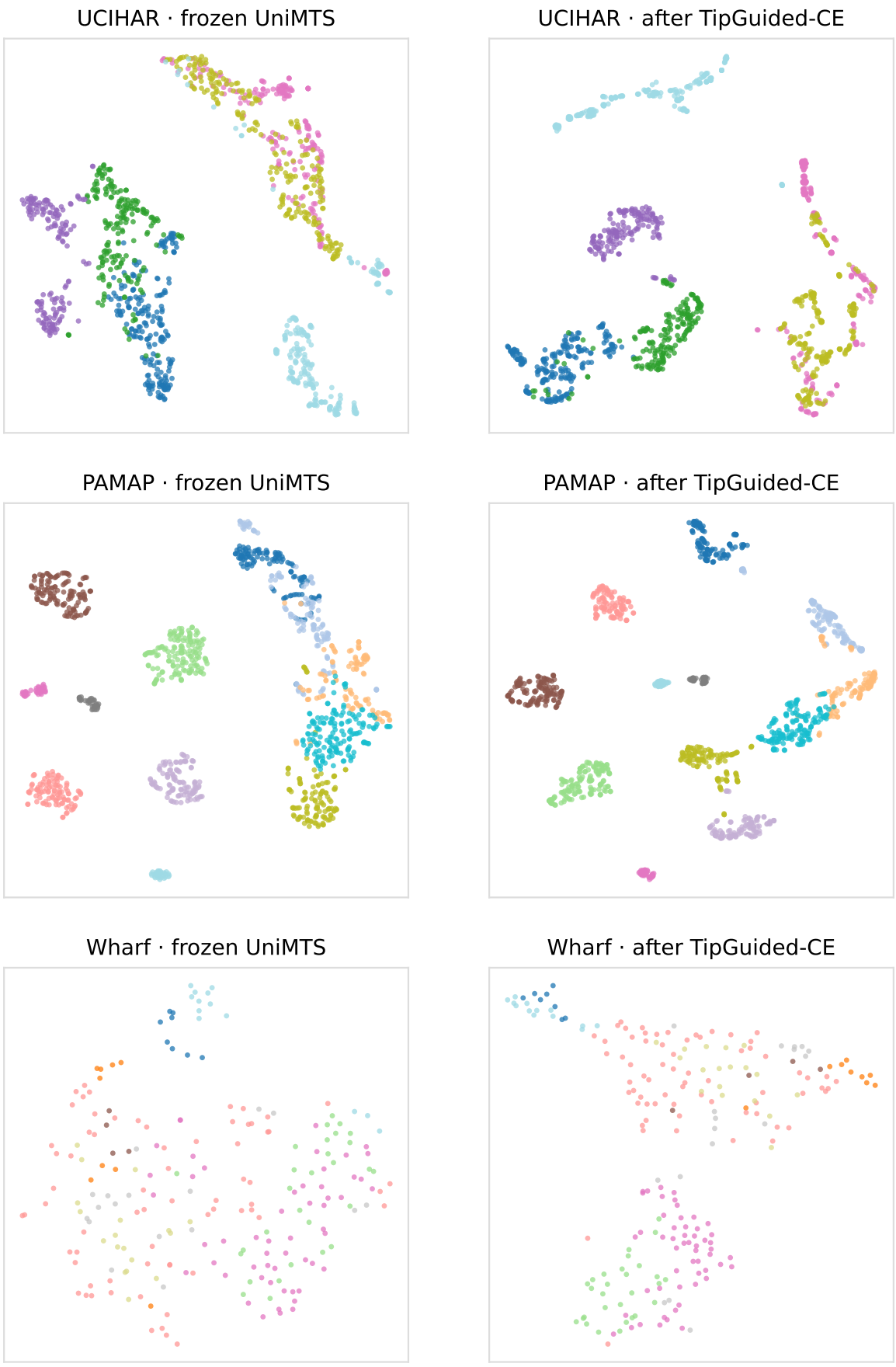


Figure 8: t-SNE of test features (k=5). Each row: frozen UniMTS (left) vs TipGuided-CE (right) for UCIHAR, PAMAP, and Wharf. TipGuided tightens class clusters and widens inter-class gaps.